Sophie Qiu

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Education

Harvey Mudd College, Claremont, California *Expected graduation May 2028*

- GPA unweighted 3.72
- Relevant Courses: Experimental Engineering, Systems Engineering, Engineering Design and Manufacturing, Chemical and Thermal Processes, Materials Engineering, Chemical Analysis, Organic Chemistry 1 and 2, Molecular Genetics

Projects

Autonomous Underwater Vehicle Spring 2026

- designed and built an autonomous underwater vehicle in 6 weeks to measure water pollution through 4 distinct sensors: turbidity, pressure, temperature, and pH
- developed custom electronic and software integrations using operational amplifiers, Arduino IDE, and MATLAB
- utilized breadboarding, 3D printing, and COMSOL simulations during the prototyping process

Zipline Optimization Simulation Spring 2026

- optimized a zipline drop height and wire tension through the application of differential equations in simulation
- applied various minimization and optimization techniques through MATLAB

Microcentrifuge Tube Closing Mechanism Fall 2026

- designed a device that reduces the time needed to close 68 1.7mL microfuge tubes by 78 percent
- communicated with client to meet project deliverables and deadlines within the given time constraint

Hammer Manufacturing Fall 2026

- operated heavy machinery such as lathes, mills, and ovens to manufacture parts within tolerances up to 5/1000th of an inch
- read and interpreted technical drawings

Work Experience

Harvey Mudd Writing Center August 2025 – Present

Consultant

- provided one-on-one writing consultation appointments for all grade levels
- reviewed and provide feedback for all types of writing including analytical essays, research theses, short stories, and internship applications

Boston Scientific June 2025 – August 2025

Operations Engineering Intern

- released formal documentation for chemical handling/disposal processes across multiple production lines
- conducted training for operators in new documentation for chemical handling/disposal processes
- created a new system for keeping track of units at different stages of the sealing and packaging process
- determined the cause of defects in rejected production units using a borescope and microscope

Cellular, Microfluidics, and Tissue Engineering Research Lab August 2024 – January 2026

Bioprinter Team

- researched chemical properties of gelatin microparticle suspension
- developed a new protocol for creating gelatin microparticles with minimally toxic reagents
- designed an adjustable bioprinter nozzle head to accommodate various needle sizes

Languages and Interests

Proficient in Mandarin Chinese.

Interests include violin, soccer, surfing, and rock climbing.